AICE3001/AICE6002: Mathematics of Machine Learning

Subsidiary Notes — Lecture 25: Entropy

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1 Keywords

- Shannon entropy, bits and nats, code length, Huffman coding, Stirling's approximation, maximum entropy

2 Main Points

2.1 Measuring Uncertainty

- Which is more uncertain: three coin tosses or one throw of a die? It depends on whether we care about the order of the tosses—but how would we answer at all?
- For a discrete random variable X, Shannon's entropy gives a precise answer,

$$H_X = -\sum_{x\in\mathcal{X}} \mathbb{P}[X=x]\,\log_2(\mathbb{P}[X=x])$$

- **A fair die**, $\mathbb{P}[D=i]=1/6$:

$$H_D = -\sum_{i=1}^{6}\frac{1}{6}\log_2\left(\frac{1}{6}\right) = \log_2(6) \approx 2.585 \text{ bits}$$

- **Three ordered coin tosses**, eight equally likely outcomes: $H_C = \log_2(8) = 3$ bits. More equally likely outcomes means more uncertainty, as we'd expect
- **Three unordered coin tosses** (we only record the number of heads): the outcomes 0 and 3 heads have probability $1/8$ each, and 1 and 2 heads have probability $3/8$ each, so

$$H_U = -\frac{1}{4}\log_2\left(\frac{1}{8}\right) - \frac{3}{4}\log_2\left(\frac{3}{8}\right) \approx 1.811 \text{ bits}$$

- This seems reasonable, but it is not obvious how we would have arrived at a number without entropy. Why this particular formula?

2.2 Additivity

- For two independent random variables, $\mathbb{P}[X,Y]=\mathbb{P}[X]\,\mathbb{P}[Y]$, so

$$\begin{aligned}
H_{(X,Y)} &= -\sum_{x,y}\mathbb{P}[X=x]\,\mathbb{P}[Y=y]\,\log_2(\mathbb{P}[X=x]\,\mathbb{P}[Y=y]) \\
&= -\sum_{x,y}\mathbb{P}[X=x]\,\mathbb{P}[Y=y]\,(\log_2(\mathbb{P}[X=x])+\log_2(\mathbb{P}[Y=y])) \\
&= -\sum_{x}\mathbb{P}[X=x]\,\log_2(\mathbb{P}[X=x]) - \sum_{y}\mathbb{P}[Y=y]\,\log_2(\mathbb{P}[Y=y]) = H_X + H_Y
\end{aligned}$$

  using $\sum_y \mathbb{P}[Y=y] = \sum_x \mathbb{P}[X=x] = 1$

- The uncertainty of two independent events adds. Shannon's entropy is one of very few functions with this property

2.3 Entropy as Code Length

- To communicate one of 2^n equally likely outcomes (say n coin tosses) we need a binary string of exactly n bits
- With 5 equally likely outcomes, 3 bits would do but would waste $3/8$ of the possible codes. Batching 3 independent messages gives $5^3 = 125$ outcomes, which fit in 7 bits ($2^7 = 128$), wasting only $3/128$: that is $7/3 \approx 2.33$ bits per message
- Batching more and more messages, the cost per message approaches $\log_2(5) \approx 2.32$ bits. In general N equally likely outcomes need $\log_2(N) = -\log_2(1/N)$ bits each
- When outcomes have different probabilities we can still find codes that use very close to $-\log_2(\mathbb{P}[X = x])$ bits for outcome x, by transmitting many messages. This code length measures the *surprise* of the outcome: rare outcomes cost more
- Shannon's entropy is the *expected code length*,

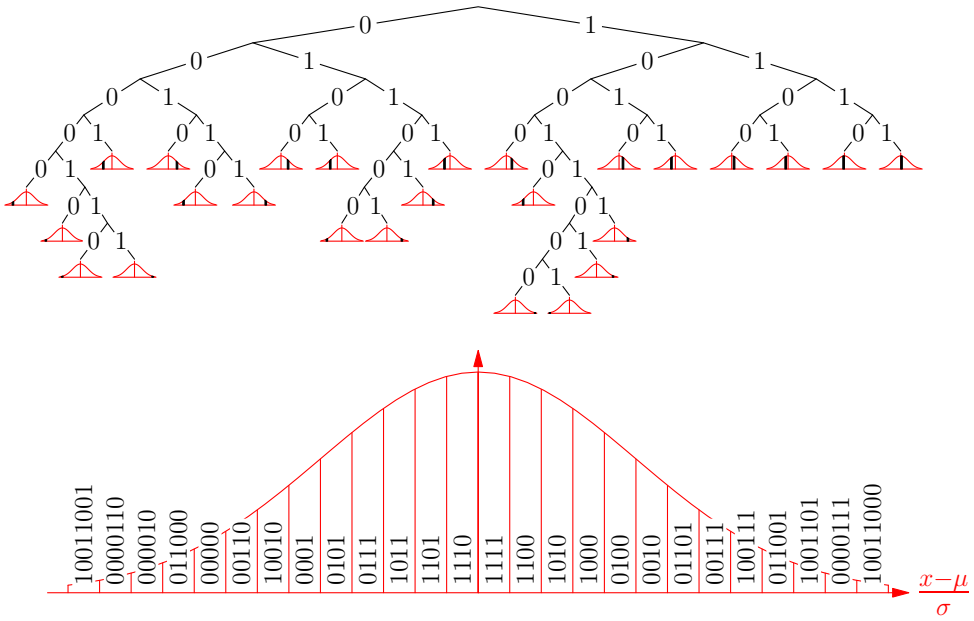
$$H_X = \mathbb{E}_X[-\log_2(\mathbb{P}[X])]$$

  which is the average amount of information needed to convey the outcome
- The lecture animation illustrates this with a loaded die: each outcome's bar is the length $-\log_2(\mathbb{P}[X = x])$ of its code word, and the entropy is the probability-weighted average bar length

X_i:	1	2	3	4	5	6
$p(X_i)$:	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{4}$
Code:	000	001	010	011	10	11
$L = -\log_2\big(p(X_i)\big)$:	3	3	3	3	2	2

2.4 Real Codes

- Code words of different lengths cause a practical problem: the receiver needs to know where each one ends
- *Huffman coding* solves this by placing the code words at the leaves of a binary tree, so no code word is a prefix of another and any message can be decoded by walking the tree
- A greedy algorithm (repeatedly merge the two least probable symbols) builds the optimal tree. Its expected code length L satisfies $H_X \leq L < H_X + 1$



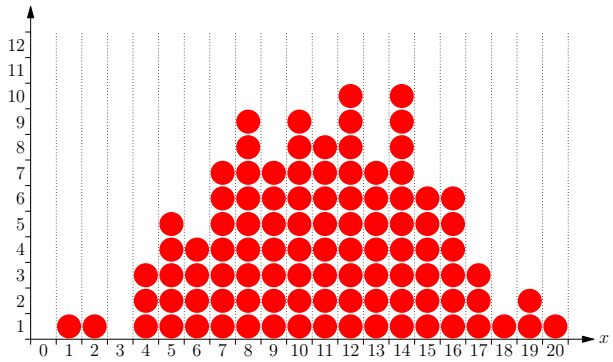
- The lecture illustrates coding a normal distribution discretised to accuracy Δx: finer accuracy means more possible values and a longer code

2.5 Bits and Nats

- With $\log_2$, entropy is measured in *bits*
- With natural logarithms, $H_X = -\sum_x \mathbb{P}[X = x] \ln(\mathbb{P}[X = x])$, entropy is measured in *nats*. One nat is $\log_2(\mathrm{e}) \approx 1.443$ bits
- Nats are more convenient whenever we do calculus on entropy, which is why the rest of this note uses natural logarithms

2.6 Counting States

- Put N balls into K boxes so that n_i land in box i

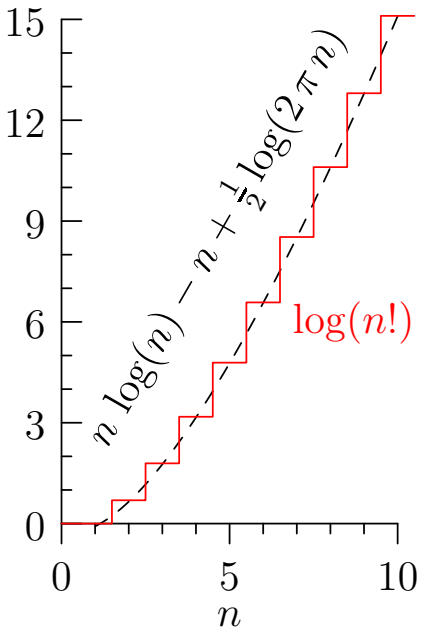


- The number of ways of doing this is the multinomial coefficient

$$W(\boldsymbol{n}) = \frac{N!}{n_1!\, n_2! \cdots n_K!}$$

- For large n, *Stirling's approximation* gives

$$n! \approx \sqrt{2\pi n}\; n^n\, \mathrm{e}^{-n} \qquad\qquad \log(n!) \approx n\log(n) - n + \frac{1}{2}\log(2\pi n)$$



- Substituting, the $-n$ terms cancel because $\sum_i n_i = N$, and to leading order

$$W(\boldsymbol{n}) = \mathrm{e}^{\log(N!) - \sum_i \log(n_i!)} \approx C\,\mathrm{e}^{N\log(N) - \sum_i n_i \log(n_i)} = C\,\mathrm{e}^{-N\sum_i \frac{n_i}{N}\log\left(\frac{n_i}{N}\right)}$$

- Writing $p_i = n_i/N$ for the fraction of balls in box i,

$$W(\boldsymbol{n}) \approx C\,\mathrm{e}^{N\,H} \qquad \text{where} \qquad H = -\sum_i p_i \log(p_i)$$

- So entropy measures the logarithm of the number of microscopic arrangements consistent with the proportions p_i. Because of the factor N in the exponent, as $N \to \infty$ the arrangements consistent with any constraints are overwhelmingly concentrated on the proportions that maximise the entropy

2.7 Entropy Is Concave

- Each term $h(p) = -p\ln(p)$ has $h''(p) = -1/p < 0$, so the entropy $H = \sum_i h(p_i)$ is a concave function of the probabilities, and $-H$ is convex
- For K outcomes, Jensen's inequality for the concave $\log$ gives

$$H_X = \mathbb{E}\left[\log_2\left(\frac{1}{\mathbb{P}[X]}\right)\right] \le \log_2\left(\mathbb{E}\left[\frac{1}{\mathbb{P}[X]}\right]\right) = \log_2(K)$$

  since $\mathbb{E}[1/\mathbb{P}[X]] = \sum_x \mathbb{P}[X = x]\,/\mathbb{P}[X = x] = K$. The uniform distribution has the most uncertainty
- Maximising a concave function subject to linear constraints (such as fixed averages) is a convex problem with a single maximum, so the maximum entropy method gives one answer
- Because $-H$ is convex and favours spread-out distributions, it is a convenient regulariser; we use it for optimal transport

2.8 The Maximum Entropy Method

- To infer a distribution from limited information, maximise its entropy subject to constraints expressing what we know. The entropy acts as a prior

- Two justifications:
  - it picks by far the most likely set of configurations consistent with the observations
  - it maximises our uncertainty given what we know, making the estimate as unbiased as possible
- It relies on the underlying arrangements being equally likely a priori. If there are other constraints we have not included, it can give poor answers
- **Known mean and variance.** For a continuous X with $\mathbb{E}[X] = \mu$ and $\mathbb{E}\left[X^2\right] = \mu_2 = \mu^2 + \sigma^2$, form the Lagrangian

$$\mathcal{L}(f) = -\int f_X(x) \log(f_X(x))\, \mathrm{d}x + \lambda_0 \left(\int f_X(x)\, \mathrm{d}x - 1 \right) + \lambda_1 \left(\int f_X(x)\, x\, \mathrm{d}x - \mu \right) + \lambda_2 \left(\int f_X(x)\, x^2\, \mathrm{d}x - \mu_2 \right)$$

- Setting the functional derivative to zero,

$$\frac{\delta \mathcal{L}}{\delta f_X(x)} = -\log(f_X(x)) - 1 + \lambda_0 + \lambda_1\, x + \lambda_2\, x^2 = 0$$

  so $f_X(x) = \mathrm{e}^{-1+\lambda_0+\lambda_1 x+\lambda_2 x^2}$
- Choosing the multipliers to satisfy the three constraints gives

$$f_X(x) = \frac{1}{\sqrt{2\pi}\,\sigma}\, \mathrm{e}^{-(x-\mu)^2/(2\sigma^2)}$$

  The *normal distribution is the maximum entropy distribution with a given mean and variance*
- Maximum entropy works superbly in statistical physics, which infers distributions over microscopic states from observations such as volume and pressure. Temperature appears, rather strangely, as a Lagrange multiplier

2.9 Softmax Is Maximum Entropy

- A model gives scores s_i to K classes. Among the probability vectors with a given average score, $\sum_i p_i s_i = \bar{s}$, which has maximum entropy? As for the normal distribution, form

$$\mathcal{L} = -\sum_i p_i \ln(p_i) + \lambda_0 \left(\sum_i p_i - 1 \right) + \beta \left(\sum_i p_i s_i - \bar{s} \right)$$

- Setting $\partial \mathcal{L}/\partial p_i = -\ln(p_i) - 1 + \lambda_0 + \beta s_i = 0$ gives $p_i \propto \mathrm{e}^{\beta s_i}$, and normalising gives the *softmax*

$$p_i = \frac{\mathrm{e}^{s_i/T}}{\sum_j \mathrm{e}^{s_j/T}}, \qquad T = \frac{1}{\beta}$$

- The temperature T is the inverse of a Lagrange multiplier, exactly as in statistical physics. It is the same "temperature" used when sampling from a language model: as $T \to 0$ the most likely class takes all the probability, as $T \to \infty$ the distribution becomes uniform
- With scores $s_i = \boldsymbol{w}_i^{\mathsf{T}} \boldsymbol{x}$ this model is multinomial logistic regression, also known as the *maximum entropy classifier*

2.10 A Little History

- Entropy was introduced in physics by Rudolf Clausius in 1865 (Macquorn Rankine discussed it in 1850)
- Ludwig Boltzmann interpreted it as counting the number of states; Josiah Willard Gibbs and James Clerk Maxwell completed the picture
- Claude Shannon founded information theory on entropy around 1948
- Ed Jaynes realised that statistical physics can be seen as an inference problem—the maximum entropy method

3 Exercises

3.1 The uniform distribution

- Using a Lagrange multiplier for normalisation, show that the maximum entropy distribution over K outcomes with no other constraints is uniform, and find its entropy

3.2 Knowing only the mean

- Find the maximum entropy density on $x \geq 0$ with a given mean $\mathbb{E}[X] = \mu$

3.3 Entropy of a biased coin

- Find the entropy $H(p)$ in bits of a coin that comes up heads with probability p. For which p is it largest? Evaluate $H(0.1)$

3.4 Finishing the normal derivation

- Show that the multipliers in the maximum entropy derivation must be $\lambda_2 = -1/(2\sigma^2)$ and $\lambda_1 = \mu/\sigma^2$

4 Answers

4.1 The uniform distribution

- Maximise $-\sum_i p_i \log(p_i) + \lambda\left(\sum_i p_i - 1\right)$. Differentiating with respect to p_i gives $-\log(p_i) - 1 + \lambda = 0$, so every $p_i = \mathrm{e}^{\lambda - 1}$ takes the same value
- Normalisation forces $p_i = 1/K$, with entropy $\log(K)$ nats or $\log_2(K)$ bits

4.2 Knowing only the mean

- Maximise $-\int_0^\infty f \log f \,\mathrm{d}x + \lambda_0\left(\int_0^\infty f \,\mathrm{d}x - 1\right) + \lambda_1\left(\int_0^\infty f\, x \,\mathrm{d}x - \mu\right)$. The stationarity condition gives $f(x) = \mathrm{e}^{-1+\lambda_0+\lambda_1 x}$
- Normalisability on $[0, \infty)$ needs $\lambda_1 < 0$; write $f(x) = A\,\mathrm{e}^{-ax}$. Normalisation gives $A = a$, and the mean of $a\,\mathrm{e}^{-ax}$ is $1/a$, so $a = 1/\mu$:

  $$f(x) = \frac{1}{\mu}\,\mathrm{e}^{-x/\mu}$$

  the exponential distribution

4.3 Entropy of a biased coin

- $H(p) = -p\log_2(p) - (1-p)\log_2(1-p)$
- $H'(p) = \log_2((1-p)/p)$, which vanishes at $p = 1/2$, where $H = 1$ bit. $H''(p) < 0$, so this is the maximum
- $H(0.1) \approx 0.469$ bits

4.4 Finishing the normal derivation

- The log density is quadratic in x, $\log f_X(x) = -1 + \lambda_0 + \lambda_1 x + \lambda_2 x^2$
- The log of a normal density is $-(x-\mu)^2/(2\sigma^2) + \text{const} = -\frac{x^2}{2\sigma^2} + \frac{\mu\, x}{\sigma^2} + \text{const}$
- Matching coefficients of x^2 and x gives $\lambda_2 = -1/(2\sigma^2)$ and $\lambda_1 = \mu/\sigma^2$. The constant, and hence λ_0, is then fixed by normalisation. This works because a normal with the right mean and variance satisfies all three constraints, and a quadratic exponent determines a normal distribution uniquely